USBSID-Pico PCB firmware manual

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Disclaimer

I do this stuff in my free time for my enjoyment. Since I like to share my joy in creating this with everyone I try my best to provide a working PCB and Firmware. I am in no way an electronics engineer and can give ***no guarantees*** that this stuff does not break or damage your hardware, computer, phone, or whatever you try to hook it up to. Be sure to take great care when inserting any real MOS SID chips into the board. While everything has been tested with real chips, this is in no way a guarantee that nothing could go wrong. Use of this board and firmware at your own risk! I am in no way responsible for your damaged hardware. That being said, have fun!

Firmware types







Each PCB revision is the firmware designator, please check your PCB revision, this is under the MOS logo and next to **USBSID-Pico** on your PCB.
Firmware filenames contain the PCB revision number e.g. `usbsidpico-rgb-v1.X.uf2` is for PCB revision `v1.0` and for use with a black clone Pico containing an RGB LED.

Firmware naming

Below are the filenames used for each Pico and board type, take great care in selecting the correct file for you Pico!
WARNING! Do *NOT* use the **RGB** firmware for any of the (non black) rp2040 or (non black) rp2350 Pico boards that do not contain an RGB LED, this can break your Pico!

Board revision

The 1.X in each filename equals for the PCB revision as mentioned under firmware types. Don't worry if you flash a firmware for the incorrect PCB revision, this causes no harm.

Firmware naming

`usbsidA-BC.uf2` examples: `usbsidpico-v1.0-rgb.uf2`, `usbsidpico2_w-v1.3.uf2` or `usbsidpico2-v1.5-rgb_em`.
A: The pico board type e.g. `pico`, `pico2`, `pico_2` or `pico2_w`

B: The PCB version e.g. v1.0, v1.3 or v1.5
C: A combination of optional features e.g. -rgb, _em or _net

Firmware naming per Pico type

Firmwares suitable for rp2040 Pico1 boards start with usbsidpico- Firmwares suitable for rp2040 Pico1_w boards start with usbsidpico_w-
Firmwares suitable for rp2350 Pico2 boards start with usbsidpico2-
Firmwares suitable for rp2350 Pico2_w boards start with usbsidpico2_w-

Naming examples for the rp2040 Pico1

Firmware naming suited for the pico 1 always start with usbsidpico-
usbsidpico-v1.X.uf2 for regular green rp2040 Pico boards.
usbsidpico-rgb-v1.X.uf2 for black clone rp2040 Pico boards with RGB LED onboard.
usbsidpico_w-v1.X.uf2 for regular green rp2040 PicoW boards.

NOTE! Highly demanding cycle-exact SID tunes (e.g. some digi/sample tunes) may not play at full speed on Pico1 or Pico1_w boards (RP2040). Use a Pico2 or Pico2_w (rp2350) board for these tunes.

Naming examples for the rp2350 Pico2

Firmware naming suited for the pico 2 always start with usbsidpico2-
usbsidpico2-v1.X.uf2 for regular green rp2350 Pico2 boards.
usbsidpico2-rgb-v1.X.uf2 for black clone rp2350 Pico2 boards with RGB LED onboard.
usbsidpico2_w-v1.X.uf2 for regular green rp2350 Pico2W boards.

Firmwares with embedded USBSID-Player and Cynthcart

Firmwares ending with em have an embedded USBSID-Player that can play PSID and RSID tunes and PSID64 PRG/P00 files and can start Cynthcart via CC commands. Both use the same embedded player core.
For example usbsidpico-v1.0_em.uf2, usbsidpico2-v1.3-rgb_em.uf2 or usbsidpico2_w-v1.5_em_net.uf2
This USBSID-Player is still in active development and some tunes might not work or play at all. The embedded player needs a lot of RAM (80% on rp2040, 40% on rp2350) and an overclocked CPU. It is built for every Pico type. On pico2_w it comes bundled with Network SID Device support (_em_net).
WARNING! Whenever the embedded player or Cynthcart is used the Pico MCU is overclocked to 250MHz. Passive cooling _might/could_ be needed.
See the USBSID-Player-manual and USBSID-Pico-Cynthcart-manual files for more information.

How to flash your Pico

NOTE:
When flashing a new firmware version, all previously configured settings will be reset to default.

Use the commandline configtool to save your current settings to a `ini` file if you want to save them!
A Raspberry Pi Pico board is incredibly easy to flash, as it comes with a built in bootloader for flashing new firmwares in the `uf2` format.
In order to flash a new firmware to your USBSID-Pico you will need to put the Pico into bootloader mode.
This can be done in multiple ways:

With the cable already plugged into your computer and Pico and seated on the board (using buttons):

- Press and hold the `BOOTSEL` button on the Pico.
- Press and release the `RST` button on the USBSID-Pico board.
- Now release the `BOOTSEL` button.
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

With the cable not plugged in to your computer and Pico and also not seated on the board (using buttons):

- Plug in the USB cable to your Pico and not into your computer.
- While holding the `BOOTSEL` button on the Pico plugin the other end of the USB cable into your computer.
- Now release the `BOOTSEL` button.
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

With the cable already plugged into your computer and Pico and seated on the board (using software):

- With the GUI config tool
- Go to the `DEBUG` tab and click on `BOOTLOADER`
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

or

- Go to the config webpage (usbsid.loudai.nl/index.html?player=webusb&debug=usbsidpico):
- Connect to the board using `WebUSB (Hermit)` as emulator

- Click on `CONFIG` to open the configuration tab
- Scroll to the bottom of the page and click on `BOOTLOADER` in the box named `DEBUG`
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

or

- With the CLI config tool command in your `PATH` variable
- Type `cfg_usbsid -boot` to send the reboot to bootloader command
- A new drive should appear on your computer called `RPI-RP2` or `RP2350` depending on your Pico type.
- Copy the correct `uf2` firmware file to this directory.
- After copying the Pico will reboot and your Pico is flashed.

Firmware configuration

USBSID-Pico has an extensive set of configuration options that expands with each firmware release.

Drivers

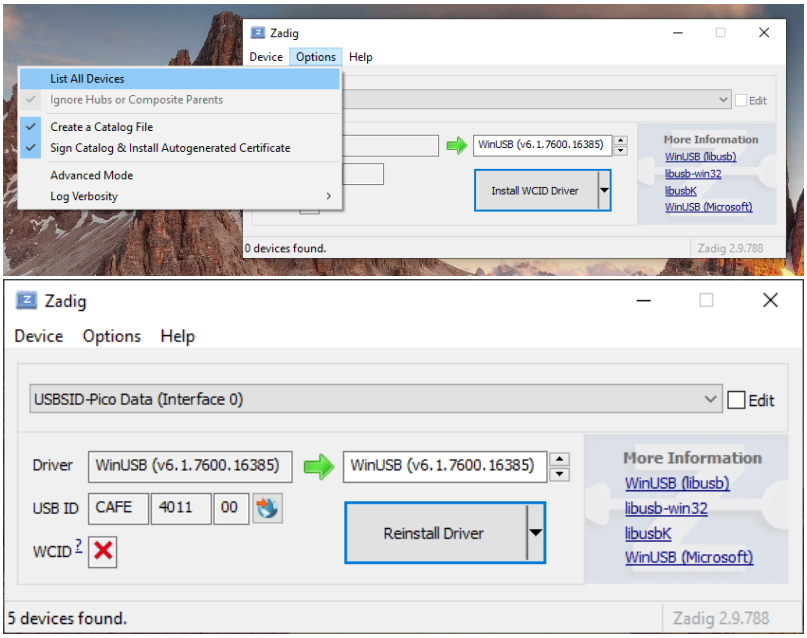
Linux Udev rules

In the examples/udev directory you can find the udev rules that are used on Linux.
This purely an example file that you can use and change to your own needs. Steps required for this to work

```
# Check if you are in the plugdev group with the groups command
groups
# If not, add yourself to the plugdev group, then log out and in
sudo usermod -aG plugdev $USER
# Copy the udev ules file to the correct directory
sudo cp 69-usbsid-permissions.rules /etc/udev/rules.d
# Now reload the udev rules
udevadm control --reload-rules && udevadm trigger
# Not working? Try reloading the service
sudo systemctl restart udev
```

Windows driver

Use Zadig to install the correct driver. Replace the driver for `USBSID-Pico Data` with WinUSB.



USBSID-Configtool

You can configurate your board by using the official USBSID-Configtool
The tool is available at USBSID-Configtool in the releases section

Commandline configuration tool

You can configurate your board with: - the commandline config-tool - the commandline config-tool with the Python Configuration GUI by ISL/Samar if you prefer a GUI.

Python Configuration GUI downloads

Download script
Python3 → Download
TkInter → Installation guide
And either one of the Linux or Windows CLI downloads below

Linux / Windows downloads

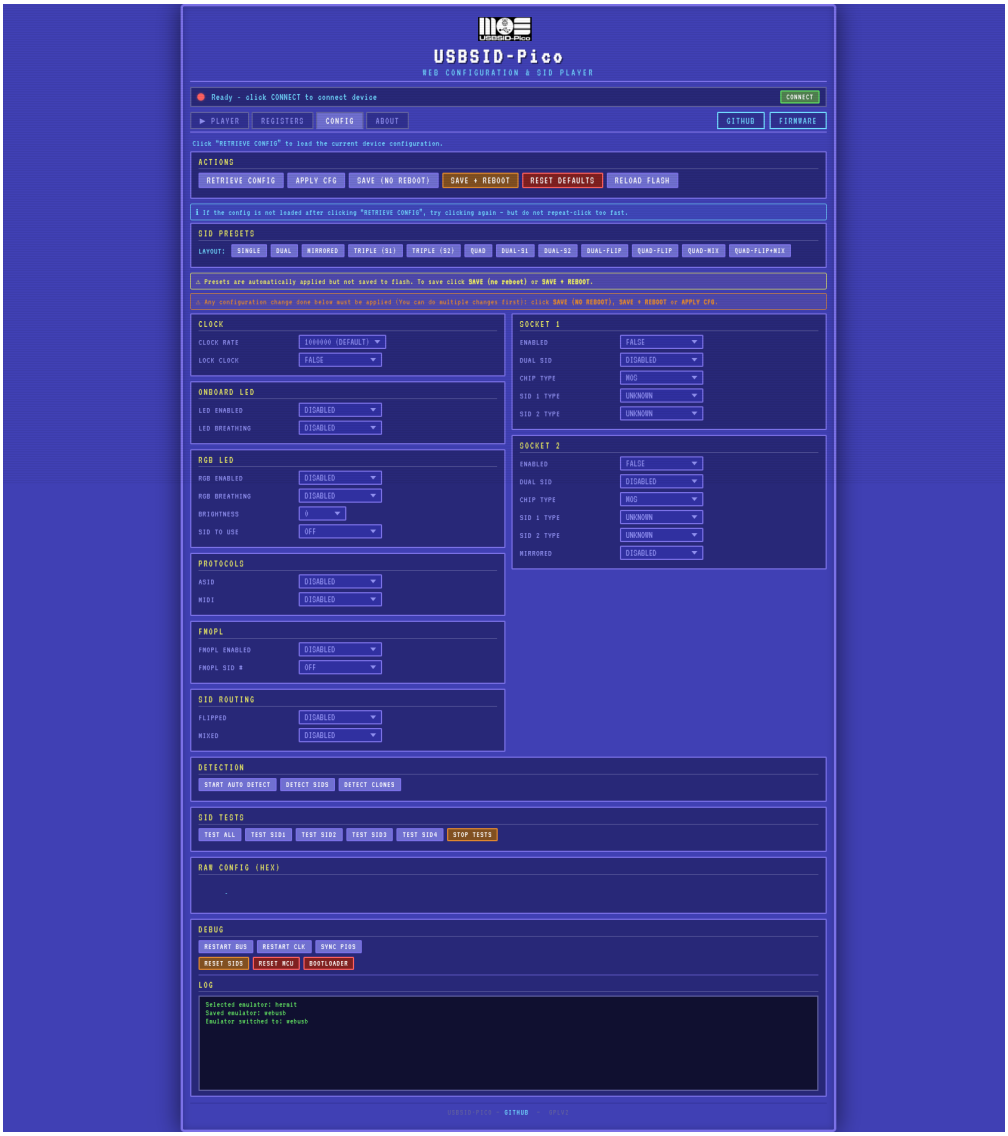
The latest version of the CLI config tool is available in a zip file called `tools.zip` from release v0.6.0-BETA and up

Windows downloads

If the DLL's are missing from the zip file you can always download them from here:
Download DLL 1
Download DLL 2

Configuration web page (for Chrome/Edge based browsers using webusb)

USBSID web configuration tool (requires a Chrome based browser).
If needed you can change your USBSID configuration after selecting WebUSB and clicking on `Open config`.



The player is set up with borrowed code from Deepsid using Hermit's JsSID implementation.

Default configuration (depends on the board version)

The default configuration differs slightly between Pico1 (rp2040) and Pico2 (rp2350) boards.
An explanation for these settings can be found further on in this document.
Board clock rate: `1000000Hz (DEFAULT)` for rp2040, `985248Hz (PAL)` for rp2350
Clock rate locked: `False`
Audio mode: `Mono`
Preset silent auto detection: `False` (`True` for PCB revision v1.5+)
Socket One Enabled: `True`
Socket One Dualsid enabled: `False`
Socket One Chiptype: `Unknown`
Socket One SID 1 type: `Unknown`

Socket One SID 2 type: `N/A`
Socket Two Enabled: `True`
Socket Two Dualsid enabled: `False`
Socket Two Chiptype: `Unknown`
Socket Two SID 1 type: `Unknown`
Socket Two SID 2 type: `N/A`
Socket Two Act as One: `False`
Flip Socket One and Socket Two: `False`
Mix socket addresses (Quad SID): `False`
LED Enabled: `True`
LED Idle breathe enabled: `True`
RGBLED Enabled: `False` (`True` for RGB Pico's)
RGBLED Idle breathe enabled: `False` (`True` for RGB Pico's)
RGBLED Brightness: `0` (`127` for RGB Pico's)
RGBLED SID to use: `-1` (`1` for RGB Pico's)
CDC Enabled: `True`
WebUSB Enabled: `True`
Asid Enabled: `True`
Midi Enabled: `True`
USB RX ringbuffer: `True`
WiFi Enabled: `False` (`_w` boards only)
NSD Enabled: `True` (`_w` boards only, needs WiFi Enabled)
Discovery Enabled: `True` (`_w` boards only)
Bluetooth NSD Enabled: `False` (`_w` boards only)
NSD port: `6581` (`_w` boards only)
Hostname: `USBSID-Pico` (`_w` boards only)
FMOpl Enabled: `False`
FMOpl SID no: `0`

Board clock rate

The Commodore64 has different clockrates depending on where you live, USBSID-Pico supports these different clock rates as configuration setting.
Supported SID players will automatically send a new clock rate request to the board to switch to.
Available options are:
`DEFAULT 1000000Hz`, `PAL 985248Hz`, `NTSC 1022727`, `DREAN 1023440`, `NTSC2 1022730`

Clock rate locked

Some players start sending data directly after the clockrate is changed. While in most cases this is no issue, some SID clones need some time to start up. This causes issues when the clock rate is changed on the fly.
Also changing the clock rate multiple times after another can cause data corruption to the SID which causes garbled sound.
To overcome this it is possible to lock the clockrate to the current setting it is set to. This blocks the ability for on the fly change requests.
Available options are:

`False`, `True`

Audio mode

PCB revision v1.3 only
Sets the audio mode for the headphone jack and audio out pins.
When set to `Mono` the audio of both SID sockets will be combined to mono.
When set to `Stereo` the audio of both SID sockets will be separated into left for `SID1` and right for `SID2`
Available options are:
`Mono`, `Stereo`

Lock audio switch

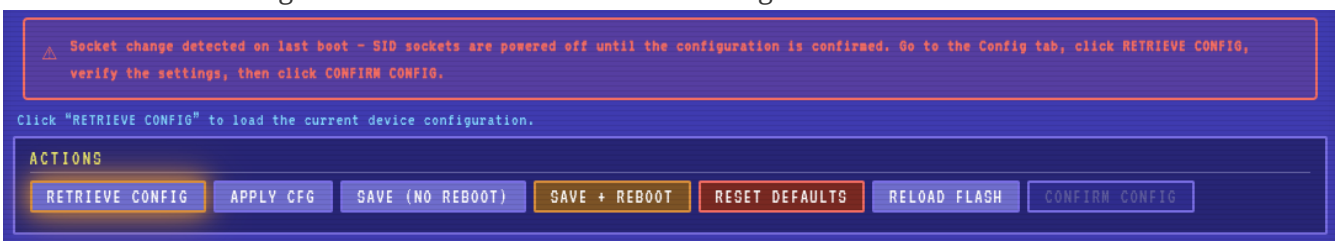
PCB revision v1.3 only
Locks the audio switch into its current state (`Mono` or `Stereo`) and prevents it from being changed.
Available options are:
`False`, `True`

SID Voltage & Configuration confirmation

PCB revision v1.5 only
The board boots with 5v and 9v enabled to automatically detect any changes in the socket / chip configuration, it does this by running a chip and SID detection.
When a change is detected the boards turns off the 9v and 5v and the onboard (RGB)LED starts flashing, making the board unusable until the configuration is validated and confirmed.
Please verify the configuration with care before confirming, failure to verify might damage/kill your SID chips, and that must be avoided!
There are multiple ways to validate and confirm the configuration:
GUI Configtool
Opening the configtool and connecting to the board shows a text stating the configuration needs to be confirmed. Verify the config on the `SOCKETS` tab and when done click `CONFIRM`



Web Configtool
Opening the config website and connecting to the board shows a text stating the configuration needs to be confirmed. Verify the config on the `CONFIG` tab by clicking `RETRIEVE CONFIG` first and when done validating click `CONFIRM CONFIG`. Both buttons glow when a click is needed



CLI Configtool

First read the configuration with `cfg_usbsid -r`, the printed configuration has a line stating the config needs confirmation. Confirm with `cfg_usbsid -ack`.

Disable socket change detection

PCB revision v1.5 only
When you have a fixed socket setup, e.g. you do not plan on changing the chips anytime soon, it is possible to disable the change detection on boot. The board then starts up with 5v and 9v and, if 12v is required, switches the voltage back off, sets the correct hv voltage and switches it back on during boot.
When the socket detection *is* on, the board will *always* try to detect any changes in the configuration *before* the required voltage is applied.
By setting the `Disable socket detection on boot` option you can control the above behaviour:
GUI Configtool



Web Configtool



CLI Configtool
Change the behaviour with `cfg_usbsid -sad N` where N can be 0 (disabled) or 1 (enabled)

Socket settings

Each socket comes with its own set of configurable options.

Socket One/Two Enabled

Enable or disable the use of this socket.
Available options are:
`False`, `True`

Socket One/Two Dualsid enabled

Enable or disable dualsid for this socket.
This requires the `Chiptype` for this socket to be set to any clone chip (not `MOS`) and ofcourse a dual SID supporting SID clone with support of connecting the A5 address line.
Available options are:
`False`, `True`

Socket One/Two Chiptype

Set the type of SID chip for this socket.
Available options are:
`MOS` (real SID), `Unknown`, `SKPico`, `ARMSID`, `ARM2SID`, `FPGASID`, `RedipSID`, `PDSID`, `BackSID`, `SIDEmu`

Socket One/Two SID 1 type

Available options are:
`Unknown`, `N/A`, `8580`, `6581`, `FMOpl`

Socket One/Two SID 2 type

Available options are:
`Unknown`, `N/A`, `8580`, `6581`, `FMOpl`

Socket Two Act as One

When enabled, this will disable other Socket Two settings and this socket will receive all writes from Socket One as it were mirrored.
Available options are:
`False`, `True`

Flip Socket One and Socket Two

When enabled, Socket One and Socket Two are swapped: Socket One's physical position responds as Socket Two and vice versa.
Available options are:
`False`, `True`

Mix socket addresses (Quad SID only)

When enabled, addresses for a 4-SID setup are mixed between sockets and not assigned sequentially per socket. Combine with `Flip` for the remaining address orderings.
Available options are:
`False`, `True`

Preset silent auto detection

When enabled, the board silently re-runs chip/SID auto detection before applying a preset.
Default is `False` on PCB revision v1.0~v1.3 and `True` on PCB revision v1.5+.
Available options are:
`False`, `True`

LED Enabled

Disable or enable the use of the onboard LED.
When enabled this automatically enables the LED as Vu meter for the voices of `SID1`.

Available options are:
`False`, `True`

LED Idle breathe enabled

Disable or enable the idle breathing effect for the onboard LED.
Available options are:
`False`, `True`

RGBLED Enabled

Only for black Pico clones with onboard RGB LED
Disable or enable the use of the onboard RGBLED.
When enabled this automatically enables the RGBLED as Vu meter.
Available options are:
`False`, `True`

RGBLED Idle breathe enabled

Only for black Pico clones with onboard RGB LED
Disable or enable the idle breathing effect for the onboard RGBLED.
Available options are:
`False`, `True`

RGBLED Brightness

Only for black Pico clones with onboard RGB LED
Set the maximum brightness of the onboard RGBLED.
Available range is:
`0 to 255`

RGBLED SID to use

Only for black Pico clones with onboard RGB LED
Set the SID of which to use the voices for the Vu meter.
Available options are (depending on your SID configuration):
`-1` off, `0` off, `1` SID1, `2` SID2, `3` SID3, `4` SID4

FMOpl Enabled

Requires a SID clone with FMOpl capabilities, the socket `Chip type` set to `Clone` and `SID type` set to `FMOpl` before activation!
Available options are:
`False`, `True`

FMOpl SID number

This setting cannot be changed manually This setting will automatically be set to the correct SID number when all requirements are met as described under `FMOpl Enabled`

CDC Enabled

This config setting is intended for future use Available options are:
`False`, `True`

WebUSB Enabled

This config setting is intended for future use Available options are:
`False`, `True`

Asid Enabled

Disable or enable ASID protocol handling (SysEx-based SID playback over MIDI).
Available options are:
`False`, `True`

Midi Enabled

Disable or enable MIDI input handling (note/CC/voice control of the SID(s)).
Available options are:
`False`, `True`

USB RX ringbuffer

The firmware keeps up to 8 incoming USB packets (64 bytes each, USB OUT transfers from host to device) in a ring and plays them from the main loop, USB keeps receiving between SID writes.
Without the ring, one packet is playing and one is waiting in the USB controller, which is about 160 us of writes in hand at 94k writes/s. A host USB scheduling delay longer than that (busy USB bus, mouse) leaves the SID idle, which is audible as a crack on write-heavy tunes (digis, e.g. Justinblue_tune_6 at about 94k writes/s).
With the ring about 1.4 ms of writes are in hand at that rate.
Applies to the CDC and WebUSB (vendor) interfaces and therefore to all host software (usplayer, VICE, sidplayfp, web player).
Default is `True` on all boards and Pico types (rp2040 and rp2350). When set to `False` the original receive path is used.
The change applies after a reboot, the setting is latched at boot.
Notes:
MIDI and ASID input arriving while a USB packet is playing is handled right after that packet.
Reads and commands queue in arrival order behind up to 8 packets.
On rp2040 a very write-heavy tune (about 94k writes/s) plays about 0.12% behind real time (inaudible), rp2350 shows no drift.

Resource use: the ringbuffer costs about 0.6 KB RAM and about 1.5 KB flash on every board type (v1.5 firmware, measured against the same firmware without the ringbuffer). RAM figures include the reserved stack and heap, they are an upper bound.

Firmware	Flash (bytes)	Flash delta	RAM (bytes)	RAM delta	RAM of chip
pico	148988	+1464	35932	+608	13.3%
pico _em	248400	+1488	225124	+604	83.3%
pico_w	390936	+1464	38232	+604	14.1%
pico_w _net	564336	+1504	125868	+608	46.6%
pico2	144816	+1352	36704	+592	6.9%
pico2 _em	239312	+1544	226120	+588	42.5%
pico2_w	388436	+1352	39220	+588	7.4%
pico2_w _em_net	648420	+1488	316128	+588	59.4%

How to change it:
`cfg_usbsid -rxring 0` (off) or `cfg_usbsid -rxring 1` (on), the board saves and reboots when the value changes. Inside a `-w` write chain the same option sets the config value, it applies after save and reboot.
Web config: `USB RX ringbuffer` in the `USB` box, changing it saves the config and reboots the board.
Configtool: `USB RX ringbuffer` in the Features section (advanced), applies after writing and saving the config.
All three tools offer the setting on firmware newer than v0.8.1.
Protocol details for host tool authors: config field `rx_ring`, set with `SET_CONFIG` item 17 (`BOARD_RXRING`) or with the dedicated command `0x83` (`SET_RX_RING`, value 0 or 1) which saves the config and reboots, a value equal to the current setting does nothing. The setting is read back in the `READ_CONFIG` reply byte 59 (0 or 1).
Available options are:
`False`, `True`

WiFi Enabled

`pico_w`/`pico2_w` boards firmware with _net only
No GUI/Web configtool support yet, use the `cfg_usbsid` commandline configtool: `-ssid`, `-psk`, `-host`, `-wen`, `-wapply`
Enables the board's WiFi station connection. Requires a provisioned SSID and password.
Available options are:
*`False*, `True`

WiFi state and IP address

`pico_w`/`pico2_w` boards firmware with _net only
Two read-only queries report the WiFi connection:
`WIFI_STATUS` replies with 2 bytes, the link (`0` down, `1` connected) and the NSD session (`0` no session, `1`

active). A byte is 0xFF when the firmware is built without WiFi (link) or NSD (session).
WIFI_READ_IP replies with 5 bytes, the WiFi state followed by the IPv4 address, first octet first. State 0 is off (WiFi not started), 1 down (no link, waiting to connect), 2 connecting, 3 connected and 0xFF firmware built without WiFi. The address bytes are zero unless the state is 3.
The commandline configtool prints both with cfg_usbsid -wifis, and cfg_usbsid -rnc prints the IP address with the network config.

NSD Enabled

`pico_w`/pico2_w boards firmware with _net only
No GUI/Web configtool support yet, use the cfg_usbsid commandline configtool: -nsden, -nsdp, -wapply`
Enables the Network SID Device (NSD) server, a TCP listener (default port `6581) plus a UDP discovery responder. NSD clients such as ACID 64, JSidplay2 and SIDPlay find and drive the board over the LAN. Only one NSD session (WiFi or Bluetooth) is active at a time. USB has priority over the SID bus.
Changes to NSD Enabled and Discovery Enabled apply at once, no reboot needed. Disabling NSD closes the listener and ends a running TCP NSD session. A new port applies on -wapply, which also saves the network config. A listener that cannot bind its port retries every second.
The NSD server buffers writes in a ring of 4096 entries and reports BUSY to the client above about 160 ms of queued cycles (PAL), accepting writes again below about 120 ms. Read-only config queries (READ_CONFIG, READ_NETCFG, WIFI_STATUS, WIFI_READ_IP, version and feature queries and similar) do not take the SID bus.
FM OPL over NSD uses the NSD v5 addressing. A TRY_WRITE_EX or TRY_READ_EX entry at $DF40 writes the FM OPL address/index port and an entry at $DF50 the FM OPL data port of the socket set under FMOpl SID no. One entry is one port write with its delay as given. Any other $DFxx address writes nothing and keeps its delay. A client enables FM OPL with TRY_SET_FM_OPL, it is disabled on each new connection. Enabling a disabled FM OPL resets the chip and answers BUSY when MIDI or ASID holds the bus. Enabling an enabled FM OPL answers OK without a reset. USBSID-Player v1.4.0 and older do not support this addressing.
Available options are:
False, True

Discovery Enabled

`pico_w`/pico2_w boards firmware with _net only
No GUI/Web configtool support yet, use the cfg_usbsid commandline configtool: -discen, -wapply`
Enables the UDP discovery responder that answers NSD clients searching the LAN for a board. It listens on UDP port `6581 for any TCP port NSD Enabled uses. Applies at once.
Available options are:
False, True

Bluetooth NSD Enabled

`pico_w`/pico2_w boards firmware with _net only
No GUI/Web configtool support yet, use the cfg_usbsid commandline configtool: -bten`
Enables the Bluetooth Classic SPP transport for the same Network SID Device protocol as `NSD

Enabled above. No NSD client speaks Bluetooth SPP directly, existing players need a host-side RFCOMM-to-TCP bridge (see `examples/nsd_spp_bridge.py`).
Available options are:
`False`, `True`

SID register reads

Per the SID datasheet only registers `$19` to `$1C` are readable. A real or unknown chip answers a read of any other register with the value left on the data bus. That value goes to the host, the register mirror (used by the onboard emulator and the SID register stream) keeps the last written value. Clone chips keep full register readback. In mirrored mode (`Socket Two Act as One`) both sockets must hold a clone chip for this.

SID register stream (FastReads)

Optional build feature, off in the default builds
Firmware built with `-DENABLE_SID_STREAM=1` adds a second vendor USB interface (interface `7`, bulk endpoints `0x07` and `0x87`) that streams SID register frames to the host, one 63 byte frame per enabled SID per tick, up to 1000 ticks per second. Vendor control requests start and stop the stream, it stops on unmount and suspend. The build reports the feature in bit 6 of the `US_FEATURES` byte and sets `bcdDevice` to `0x0101`. Default builds have no interface 7 and `bcdDevice` `0x0100`.
The frames are copied from the board's register mirror, the stream does not read from the SID chips.
Batch reads are part of every build and do not need this option. One `READ` packet with a register count reads up to 63 registers. Hosts detect it by firmware version 0.8.1 or newer.

Debug logging

The CDC and vendor (WebUSB) event logs (mount, line state, line coding, break, vendor TX and control requests) print only when the build has USB IO debugging enabled (`USBIO_DEBUGGING` in `CMakeLists.txt`, default `0`). Earlier builds always printed them over UART, each log line stalled playback for milliseconds.

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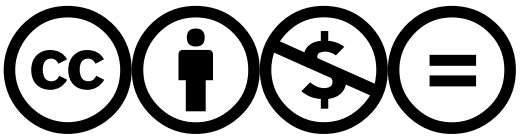
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